

Windows Event Investigation Report

Report Information

Report title: Windows Event Investigation

Lab number: Lab 001

Analyst: Giovanni Rivera

Environment: Personal Windows 11 lab computer

Investigation type: Training exercise

Status: Completed

Executive Summary

A beginner-level investigation was conducted using Windows Event Viewer to examine System and Security logs.

The investigation reviewed a DistributedCOM Warning with Event ID 10016 and a successful logon event with Event ID 4624.

The 10016 event was determined to be a common Windows warning that did not independently indicate malicious activity.

The 4624 event involved the SYSTEM account and a computer account ending with a dollar sign. Based on the available evidence, the activity was most consistent with normal Windows internal operations.

No evidence of malicious activity was identified during this limited investigation.

Investigation Objective

The objective was to learn how to locate Windows events, identify basic event metadata, search for a specific Event ID, and make a cautious evidence-based assessment.

Evidence Reviewed

1. Windows System log

2. DistributedCOM Event ID 10016
3. Windows Security log
4. Successful logon Event ID 4624
5. Subject account information associated with the logon

Timeline

7/31/2026 at 7:46:06 PM

A DistributedCOM Warning with Event ID 10016 was recorded in the Windows System log.

7/31/2026 at 8:13:50 PM

A successful logon event with Event ID 4624 was recorded in the Windows Security log.

The event included the SYSTEM security identity and a computer account ending with a dollar sign.

Findings

Finding 1:

Event ID 10016 was categorized as a Warning.

A Warning identifies activity that Windows considered worth recording, but it does not automatically identify a cybersecurity incident.

Finding 2:

Event ID 4624 records a successful logon.

The reviewed event involved SYSTEM and a computer account rather than clear evidence of an interactive human logon.

Finding 3:

A single event is insufficient to prove malicious activity.

Additional context would be required, including the logon type, associated processes, source address, nearby events, and whether the activity was expected.

Analyst Assessment

Classification: Expected or likely benign system activity

Confidence: Moderate

Reasoning:

The account was identified as SYSTEM, and the account name ended with a dollar sign, which is commonly associated with a computer account. There was no additional evidence in this limited review showing unauthorized access or malicious execution.

Limitations

This investigation reviewed only a small selection of Windows events.

Sysmon, endpoint detection tools, centralized SIEM logs, process telemetry, network telemetry, and threat intelligence were not yet available.

Because of those limitations, this report does not prove that the entire computer is free from malicious activity. It only assesses the specific events reviewed.

Recommended Next Steps

1. Learn how to interpret Windows logon types.
2. Investigate failed logon events.
3. Review account creation and permission-change events.
4. Install Sysmon to collect detailed process information.
5. Connect the endpoint to a SIEM for centralized analysis.
6. Build a larger timeline using related events.

Final Conclusion

The reviewed events did not provide sufficient evidence of a cybersecurity attack.

The activity was most consistent with routine Windows system behavior.

The primary lesson from the investigation was that analysts must evaluate account information and surrounding context rather than treating every Warning or successful logon as malicious.

Guided-Lab Disclosure

This investigation was completed through guided beginner instruction. I personally performed the Event Viewer navigation, log searches, observations, reasoning exercises, evidence collection, and documentation.

Personal Reflection

Before this Lab I was a very new beginner to learning anything cybersecurity or IT. I am also learning in the google cybersecurity course about 20% through preparing for the security + exam. The most confusing part was understanding how to audit something that is constantly adding new logs faster than you can observe, and proper protocol to observe. Understanding the Event ID and learning how to find specific logs for it became clearer to navigate after being guided. It also became clearer how to think and approach a situation like warning level v.s error level in the system windows logs. One event like a 4624 Logon is not enough to prove an attack because it can be a variety of factors. Looking into the Security ID and Account Name I can find clues to conclude whether it is a human, or windows itself performing system tasks. I would like to investigate more Logs and see activity updates in real time and learn how to properly spot possible risks in the activity of an organization. The step I could explain easily to another beginner is the process of getting into the event viewer looking into the windows logs and opening security logs to look for recent logons by clicking "find" and searching for the ID 4624.